ARYAN GUPTA

Applied Physics / Scientific Computing / Research Software / Engineering AI

Email · Portfolio · GitHub · LinkedIn · Google Scholar · ORCID

PROFESSIONAL PROFILE

Applied physicist and engineer with three years of industry experience spanning first-principles carbon-capture R&D, molecular and continuum modeling, instrumentation and controls, and engineering AI workflows. Builds reproducible computational research, scientific evaluation methods, and software systems across inverse problems, nonlinear dynamics, scientific ML, robotics, and LLM infrastructure. Currently pursuing an MSc in Applied Computational Science & Engineering at Imperial College London.

EDUCATION

Imperial College London | London, UK Sep 2026 – Sep 2027 (expected)

MSc, Applied Computational Science & Engineering | In progress

University of Michigan | Ann Arbor, MI, USA Sep 2019 – May 2023

BSE, Engineering Physics, cum laude | Minors: Electrical Engineering, Computer Science, Entrepreneurship

INDUSTRY EXPERIENCE

Technip Energies | Houston, TX Aug 2023 – Jul 2026

Positions and project assignments overlapped; individual periods are recorded below.

Technology Developer I | Applied Physics / Carbon-Capture R&D Aug 2023 – Jul 2025; Feb–Mar 2026

- Established the team's first-principles molecular modeling capability, using molecular dynamics (MD) and density functional theory (DFT) across high-performance computing and cloud systems to support carbon-capture materials/process R&D.
- Linked atomistic model outputs with diffusion and computational fluid dynamics (CFD) workflows and experimental interpretation, connecting material-scale mechanisms to higher-level transport questions.
- Cut simulation wall time by approximately two-thirds through GPU/cloud execution, parallel I/O, and workflow improvements; automated parameter sweeps and standardized repeatable computational runs.
- Developed analysis workflows for terabyte-scale image data, reducing formerly multi-day manual analysis tasks to hours; incorporated particle characterization and uncertainty-aware analysis.
- Built surrogate-modeling, active-learning, numerical optimization, and verification workflows to help prioritize simulations or experiments and test model behavior before use.
- Authored 15 reviewed internal technical notes and a roughly 70-page market/business analysis used in funding and roadmap discussions; maintained reusable technical assets and mentored colleagues.

Americas AI Solutions Initiative | Co-Founder, Co-Lead & Technical Lead Dec 2025 – Jul 2026

- Co-founded and co-led an internal engineering AI initiative across four Americas operating centers and eight engineering functions; scoped and piloted a portfolio of five AI/data workflows while concurrently serving in an engineering position.
- Set the initiative's technical direction and reusable architecture: engineering data retrieval, structured-query generation, document extraction, multimodal document understanding, validation, and human-review mechanisms, rather than isolated one-off demos.
- Led development of a Python LLM-assisted natural-language-to-SQL pipeline across three engineering datasets, using SQL-backed retrieval and query-generation/evaluation logic to make structured engineering information accessible through natural-language questions.
- Coordinated an LLM-assisted vendor bid-tabulation workflow that parsed lengthy, unstructured specification packages and extracted, normalized, and compared requirement-level information; reduced manual review effort by approximately 80% in the evaluated workflow.
- Built and coordinated a local-first P&ID intelligence workflow for extracting instruments, lines, loops, labels, connectors, crossings, and cross-page relationships from engineering drawings; combined PDF/vector parsing, OCR, computer vision/object detection, rules, validation, and engineer-facing review outputs.
- Developed an internal EPS benchmarking and proposal-analytics desktop tool for historical estimating packages: recursive document scanning, PDF field extraction, incremental SQLite caching, revision selection, source catalog, and explainable review queues.
- Extended the EPS tool with stakeholder-facing filtering and comparison views, KPI/summary displays, alias normalization, preserved manual overrides, portable workspaces, and Excel exports; refined usability and data quality from direct estimator and proposal-team feedback.
- Established practical MVP delivery and assessment methods: engineering stakeholder interviews, agile sprint reviews, design-verification-fitness checks, baseline comparisons, evaluation criteria, Pugh decision matrices, technical roadmaps, and handoff documentation.
- Coordinated cross-disciplinary contributors and partnered with discipline managers and global digital stakeholders to align projects, avoid duplicated work, share reusable components, and identify paths for wider adoption.
- Presented the initiative and its project portfolio at internal events, town halls, and leadership forums, generating visibility and support for subsequent engineering AI pilots.

Specialist Engineer I | Instrumentation & Controls Jul 2025 – Jul 2026

- Worked on hydrogen, carbon-capture, and brownfield engineering scopes, reviewing instrumentation and control design deliverables against project requirements.
- Cross-checked P&IDs, process flow diagrams, cause-and-effect diagrams, control narratives, vendor documentation, and specifications for consistency and traceability.
- Gained experience with standards-driven engineering QA, documentation control, revision management, and cross-discipline interfaces in conventional project delivery.

ADDITIONAL CONTRACT ENGINEERING EXPERIENCE

Assistive Mobility Robotics Startup | Project-Based Engineering Consultant Summer 2026

External client engagement; preliminary engineering feasibility and sizing, not an independently owned product.

- Developed a C++20 analysis tool to screen a mobility platform's quasi-static support-polygon stability, center-of-mass margins, friction/slip limits, and support-transition scenarios across bounded design and environmental conditions.
- Estimated sagittal hip/knee force and torque envelopes using inverse kinematics and Jacobian statics; evaluated sensitivities, limiting cases, and adverse parameter combinations to identify early design risks.
- Produced evidence-linked feasibility and sizing recommendations with explicit assumptions, contact-model limitations, and verification needs; did not claim dynamic-system or hardware validation.

INDEPENDENT RESEARCH & PUBLICATIONS

Research papers, preprints, and associated reproducibility artifacts, with dissemination status and supporting links recorded for each work.

Trajectory-Only Structural Regime Detection in Deterministic Dynamical Systems Feb – May 2026

Preprint + reproducible Python toolkit | Preprint DOI | Code DOI | Repository

- Designed a non-ML method to detect changes in deterministic nonlinear dynamics using only multivariate trajectories and ordered parameter sweeps; withheld governing equations and oracle regime labels from the detector.
- Combined trajectory-derived indicators into a structural score and extracted candidate boundaries, then compared them post hoc against separately defined qualitative reference boundaries.
- Evaluated FitzHugh–Nagumo and forced van der Pol systems, including nuisance-parameter tests, boundary localization, specificity checks, robustness analyses, scalar-delay-embedding supplements, and ablations.

ThermoBench-Consist v1.0 | Thermodynamic consistency diagnostics Sep – Oct 2025

Preprint + released research software | Documentation DOI | Code DOI | Repository

- Built a lightweight, CPU-only diagnostic suite for machine-learned thermodynamic equations of state and vapor–liquid-equilibrium (VLE) surrogates, emphasizing physically meaningful signs and slopes rather than prediction error alone.
- Implemented checks for density–pressure monotonicity, positive isothermal compressibility, Clapeyron-consistent saturation curves, and speed-of-sound sanity; used reference-adapter comparisons and interpretable reports.
- Added near-spinodal and critical-region guardrails, reproducible test grids, CLI execution, JSON/HTML/Markdown output, plots, and continuous-integration checks to avoid spurious failure claims.

Assessing the Limits of GNNs for Vapor-Liquid Equilibrium May – Sep 2025

Research preprint | arXiv:2509.10565

- Investigated graph neural network surrogates for thermodynamic-property and vapor–liquid-equilibrium workflows, evaluating more than aggregate pointwise predictive accuracy.
- Documented derivative pathologies and liquid-phase inconsistencies that could make pointwise-accurate learned pressure surfaces unusable inside VLE solvers, motivating the domain-specific checks subsequently developed in ThermoBench-Consist.

Energy Efficiency in Robotics Software: A Systematic Literature Review from 2020 Onward Jan – Aug 2025

Systematic literature review + replication package | arXiv:2508.12170 | Replication DOI | Repository

- Synthesized 79 studies published since 2020 on energy-efficiency techniques and empirical evidence for robotics software, spanning autonomy workflows and software-system optimization.
- Constructed a taxonomy and reporting checklist, documented screening and extraction choices, and released frozen data and scripts to regenerate figures and tables.

RELEASED RESEARCH SOFTWARE & TECHNICAL SYSTEMS

WaveSleuth-Devito | Inverse Wave Playground v1.0.0 Jun – Aug 2026

Released inverse-problem sandbox | Repository | v1.0.0

- Built a Devito-backed, two-dimensional acoustic inverse-problem sandbox combining deterministic wave propagation, sparse receiver traces, configurable acquisition geometries, and hidden synthetic media.
- Implemented inspectable circle/ellipse/grid-mask reconstruction baselines, mismatch objectives, open and blind challenges, active source selection, uncertainty summaries, physical and budgeted scoring, and release validators.
- Canonical v1.0 toy challenge scores reached IoU 1.000 for a mask, 0.895 for a joint circle-radius/velocity case, 0.800 for a circle, and 0.787 for an ellipse; these are deliberately synthetic benchmarks, not field-scale validation.

- Packaged CLI and testing workflows for reproducibility, including simulation, inversion, visualization, challenge-suite generation, diagnosis, and report validation.

Costgate v1.0 | LLM inference-cost regression gate 2026

Open-source CI / model-evaluation tooling | Repository | Code DOI

- Developed a local- and CI-first tool for detecting inference-cost regressions between a baseline and candidate LLM run over identical task suites.
- Separated provider/API success from task-valid success; measured cost per valid success alongside token consumption, latency, task-success rate, and run-to-run variability.
- Implemented schema-versioned result artifacts, configurable rate cards, provider abstraction, replay/mock testing, suite validators, baseline-family compatibility checks, and policy-based pass/warn/fail outcomes.
- Supported practical and statistical regression gates, including minimum absolute changes and nonparametric significance tests; integrated with GitHub Actions and machine-readable/Markdown reports.

REB-1: Robot Energy Benchmark v0.1 Aug 2025

Released robotics benchmarking utility | Code DOI | Repository

- Built a ROS 2-friendly power and thermal logging micro-benchmark that records power traces, integrates watt-hours, and estimates gCO_2 using user-specified grid-intensity assumptions.
- Provided nvidia-smi or deterministic demonstration-data sources, optional /cmd_vel activity overlays, a fixed tidy-CSV schema, analysis notebook, reproducible example data, tests, and CI that runs without ROS installed.

EARLIER RESEARCH & TEACHING

BIRDS Lab, University of Michigan | Research Assistant 2021 – 2023

- Contributed to an adaptable quadcopter test platform, spanning embedded hardware, ESP32/FTDI Wi-Fi/UART telemetry, power electronics, flight-control software, and mechanical/electrical integration.
- Debugged cross-subsystem behavior and implemented iterative hardware/software modifications for flight-platform integration.

EECS 314: Circuit Design & Analysis, University of Michigan | Instructional Aide 2022 – 2023

- Led discussion groups for approximately 80 students and a 20-student circuits laboratory covering circuit analysis, RLC transients, operational amplifiers, and bench instrumentation.
- Supported course delivery and instructional coordination for more than 160 students.

MiTEE Nanosatellite Project | Plasma Team / Structures Liaison 2020

- Worked on electrodynamic-tether propulsion feasibility for pico/nanosatellites, including thermodynamic analysis, electronics prototyping, and the interface between plasma and structures subsystems.

ENTREPRENEURSHIP & ENGINEERING BUILDS

HelixCases | Co-Founder & Software Developer 2022

- Co-built a university technical-recruiting competition platform reaching more than 550 students at 20 universities; supported platform development, product scoping, customer discovery, and event operations.
- Secured a $10,000 Lyft contract for the platform/competition effort.

Electric Longboard Remote Controller | Embedded systems 2022

- Designed and assembled an STM32L010C6T6-based controller with smartphone UART/LPUART communication, battery and display feedback, and modified VESC firmware.
- Integrated embedded electronics, user controls, and a custom 3D-printed enclosure; incorporated additional smartphone/media interactions.

Out-of-Order Processor Architecture | Digital design 2022

- Implemented a MIPS R10K-style two-way superscalar, out-of-order processor architecture in SystemVerilog.
- Included instruction/data caches, load-store queue, branch prediction/recovery, precise exceptions, baseline simultaneous multithreading, and a visual debugger for bring-up.

Low-Autonomy Follower Robot | Embedded robotics & control 2022

Repository

- Developed a two-Zumo leader/follower system with STM32 Nucleo boards, LED-encoded command patterns, PixyCam vision, ultrasonic range sensing, and PID follower-distance control.

LEADERSHIP, SERVICE & MENTORING

Project RISHI | Fundraising Co-Chair 2020 – 2021

- Supported campus fundraising and grant applications; helped secure approximately $16,000 toward a $35,000 budget.

StartUM | Member & Mentor 2021 – 2022

- Mentored an early-stage medical-volunteering startup through market validation, early product reasoning, and a change in target market.

Citizens for Animal Protection, Houston | Volunteer 2023 – 2024

- Supported dog walking, kennel maintenance, and daily operations at a Houston animal shelter.

TECHNICAL SKILLS

Programming & development Python; C/C++; MATLAB/Simulink; SystemVerilog; SQL/PostgreSQL; JavaScript/TypeScript; Git; Linux/Unix; Docker; GitHub Actions / CI; reproducible packaging and testing.

Scientific computing & numerical methods NumPy; SciPy; Devito; numerical ODE/PDE methods; finite-difference modeling; inverse problems; simulation, parameter sweeps and optimization; molecular dynamics (LAMMPS); electronic structure / DFT (NWChem); uncertainty quantification.

Machine learning & evaluation PyTorch; scikit-learn; OpenCV; scientific surrogate models; active learning; statistical evaluation; benchmarking; validation and acceptance harnesses; LLM provider/latency/token/cost instrumentation.

HPC, platforms & data SLURM; GPU computing; cloud/GCP; parallel I/O; data pipelines; image/data processing; SQLite; OCR and engineering PDF/document parsing; versioned datasets and artifacts.

Embedded systems, robotics & control STM32/ARM; ESP32; FreeRTOS; UART/LPUART; power and sensing; camera geometry; PID control; actuator statics; CAD/prototyping and electromechanical integration.

Engineering domains Carbon capture; thermodynamics and VLE; diffusion / CFD coupling; instrumentation and control systems; P&IDs/PFDs; cause-and-effect and control narratives; engineering review and traceability.

HONORS & CREDENTIALS

Six Sigma Green Belt (CSSGB), Technip Energies (2026) | Foundational Finance for Strategic Decision Making, University of Michigan (2020) | BSE cum laude (2023) | University Honors and Dean's List (2019, 2020)